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MIL-C-0010597F(ME)
27 June 1994
USED-IN-LIEU-OF
MIL-C-10597E
25 August 1983

MILITARY SPECIFICATION

CLEANING COMPOUND WITH CONDITIONER

FOR ENGINE COOLING SYSTEMS

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-10597E. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers one type and grade of cleaning compound and conditioner supplied in a kit, for cleaning and conditioning liquid cooled, internal combustion engines. No military symbol is recorded.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6850

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SPECIFICATIONS

FEDERAL

O-O-690	- Oxalic Acid, Dihydrate, Technical.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-591	- Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging.
MIL-B-26701	- Bottles, Screw Cap and Carboys Polyethylene Plastic.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipping and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 171 - 179 - Hazardous Materials Regulation.

(Application for copies should be addressed to: SUPT OF DCMNTS, US GOVT PRINTG OFC, WASHINGTON DC 20402.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3400	-	Standard Specification for Liquid Sodium Silicates.
E 29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to: NATL MTR FRGHT TRFC ASS, ATTN TRFC ORDR SECT, 2200 MILL RD, ALEXANDRIA VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to: UNFRM FRGHT CLASS CMMTE, ATTN TARIFF PBLSHNG OFCR, ROOM 1106, 222 S RIVERSIDE PLZ, CHICAGO IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), samples shall be subjected to first article inspection (see 6.3) in accordance with 4.5.

3.2 Materials. Materials shall be as specified herein.

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3.2.1 Recovered materials. For the purposes of this requirement, recovered materials are those materials which have been collected from waste materials and reprocessed to become a source of raw materials. The components used in the cleaning compound specification may be newly fabricated from recovered materials, provided the resultant product produced meets all the requirements of this specification.

3.3 Cleaner.

3.3.1 Composition. The cleaner shall consist of two components. The first component shall be oxalic acid as specified in table I. The second component shall consist of an aqueous solution of aluminum chloride as specified in table I.

TABLE I. Cleaner composition.

Components	Requirements		Applicable test paragraph
	minimum	maximum	
Oxalic acid, dihydrate, O-O-690, class II, grams	287.5	295.0	4.7.2.1
Aluminum chloride solution, grams	49.0	53.0	4.7.2.2

3.3.2 Cleaner chemical requirements. The aluminum chloride solution shall conform to the requirements of table II.

TABLE II. Cleaner chemical requirements.

Components	Requirements		Applicable test paragraph
	minimum	maximum	
Aluminum chloride, anhydrous, percent by weight	22.5	28.0	4.7.2.3.1
Chlorine, percent by weight of aluminum chloride	75.0	80.0	4.7.2.3.2
pH of 1 % $AlCl_3$ solution	3.2	3.4	4.7.2.3.3

3.4 Conditioner.

3.4.1 Composition. The conditioner shall consist of two components. The first component shall consist of a mixture of trisodium phosphate, sodium borate, and sodium carbonate as specified in table III. The second component shall consist of a sodium silicate solution as specified in table III.

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TABLE III. Conditioner composition.

Components	Requirements		Applicable test paragraph
	minimum	maximum	
Trisodium phosphate, anhydrous, grams	139	150	4.7.2.4
Sodium borate, anhydrous, grams	115	125	4.7.2.5
Sodium carbonate, anhydrous, grams	50	60	4.7.2.6
Sodium silicate solution, ASTM D 3400, type 1, grams	325	400	4.7.2.7

3.5 Conditioner chemical requirements. The resultant mixture of trisodium phosphate, sodium borate, and sodium carbonate shall conform to the requirements of table IV.

TABLE IV. Conditioner chemical requirements.

Components	Maximum titer	Applicable test paragraph
Free alkali in mixture, mL of 0.1 N HCl	1.0	4.7.2.8

3.6 Toxicological product formulation. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertaining to the toxic effects shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring activity.

3.6.1 Material safety. The contractor shall comply with the material safety requirements of FED-STD-313 (see 6.4).

3.7 Transportability. Wherever and whenever applicable packaging, packing, and marking of cleaning compounds shall be prepared for delivery in accordance with the requirements of the Code of Federal Regulations 49 CFR, Parts 171-179.

3.8 Limiting values. The following applies to all specified limits in this specification: For purposes of determining conformance with these specifications, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand digit used in expressing the specification limit in accordance with the rounding-off method of ASTM E 29.

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3.9 Workmanship. Both the solid and liquid components of the cleaning compound kit shall meet all the requirements of this specification. In addition to these requirements, the solid components shall be dry, free flowing and contain no foreign matter. The sodium silicate solution may be slightly turbid in appearance, but free of insoluble suspensions such as foreign matter and undissolved sodium silicate. The aluminum chloride solution shall be clear in appearance and free of insoluble suspensions such as foreign matter and undissolved aluminum chloride.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection lot. All compounded material of the same type manufactured as one batch and offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and test.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
b. Quality conformance (see 4.6).
c. Inspection of packaging (see 4.8).

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4.4 Toxicological product formulation review. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use through the submission of the material safety data sheet detailed in FED-STD-313 (see 3.7 and 6.4).

4.5 First article inspection.

4.5.1 First article examination. The first article shall be examined as specified 4.7.1. The lot shall be rejected if one or more defects are found.

4.5.2 First article tests. The first article shall be tested as specified in 4.7.2.1 through 4.7.2.8. Failure of any test shall be cause for rejection.

4.6 Quality conformance inspection.

4.6.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.2 Examination. Samples selected as specified in 4.6.1 shall be examined as specified in 4.7.1. Presence of one or more defects shall be cause for rejection.

4.6.3 Tests. Samples selected as specified in 4.6.1 shall be tested as specified in 4.7.2.1, 4.7.2.2, 4.7.2.7, and 4.7.2.8. Failure of any test shall be cause for rejection.

4.7 Inspection procedure.

4.7.1 Examination. The cleaning compound kit shall be examined as specified herein for the following defects:

101. Material not as specified (see 3.2).
102. Workmanship not as specified (see 3.9).
103. Leakage of liquid components from bottles.
104. Leakage of dry components from bags.
105. Dry components are not free flowing.

4.7.2 Tests.

4.7.2.1 Oxalic acid, dihydrate. Weigh the bag containing the oxalic acid, dihydrate to the nearest 0.1 g. Transfer the contents to a dry stoppered flask and reweigh the bag. Record the difference in weight as the total weight of acid. Nonconformance to table I shall constitute failure of this test.

4.7.2.2 Aluminum chloride solution. Weigh the bottle containing the aluminum chloride solution to the nearest 0.1 g.

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Transfer the contents to a dry stoppered flask. Rinse the bottle five times with distilled water and once with 95 percent ethanol. Dry bottle with a stream of air and reweigh bottle. Record the difference in weight as the total weight of the aluminum chloride solution. Nonconformance to table I shall constitute failure of this test.

4.7.2.3 Cleaner chemical requirements.

4.7.2.3.1 Weight percent aluminum chloride. Weigh 4 g of the aluminum chloride solution, to the nearest 0.1 mg, into a 100-mL volumetric flask. Record this weight as the $AlCl_3$ sample weight. Dilute the weighed sample to the mark with distilled water and agitate to ensure homogeneity. Accurately pipet 10 mL of this solution to a 400-mL beaker. Add 3-5 drops of methyl red indicator, 5 g of ammonium chloride (NH_4Cl) and dilute to 100 mL with distilled water. Slowly heat this solution to a slow boil and once boiling add 6 N ammonium hydroxide (NH_4OH) drop-wise until the solution color changes to a distinct yellow. Boil solution an additional 1-2 minutes. After boiling, immediately filter hot solution through a 9-cm Whatman No. 41H filter paper or equivalent. Wash the precipitate with a 2 percent NH_4Cl solution. After washings, transfer the filter paper and its contents to a tared crucible and char over either a blast burner or a bunsen burner for 5-10 minutes. Heat at a rate such that the paper is not allowed to flame up and lose precipitate and all carbonaceous material is removed. After heating allow crucible to cool to room temperature in a desiccator. Weigh the crucible and record the weight gain as the weight of aluminum oxide (Al_2O_3). Calculate the percent aluminum chloride, anhydrous as follows:

$$AlCl_3,\ wt\ \% = \frac{Al_2O_3\ weight\ x\ 2616}{AlCl_3\ sample\ wt}$$

A precipitate color other than white or nonconformance to table II shall constitute failure of this test.

4.7.2.3.2 Weight percent chlorine. Calculate the weight percent of the aluminum chloride solution by performing test in 4.7.2.3.1 and record the value obtained. Weigh 4 g of the aluminum chloride solution to the nearest 0.1 mg, into a 100-mL volumetric flask. Record this weight as the $AlCl_3$ sample weight. Dilute the weighed sample to the mark with distilled water and agitate to ensure homogeneity. Accurately pipet 10 mL of this solution to a 400-mL beaker. Add 5 mL of 6 N nitric acid (HNO_3) and dilute to 50 mL with distilled water. Accurately pipet 20 mL

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of standardized 0.1 N silver nitrate ($AgNO_3$) solution to the sample. Stir solution during addition. On a hot plate heat solution to 90 - 95 °C until the precipitate is coagulated, then filter hot solution through a Whatman No. 44 filter paper or equivalent. Wash the precipitate with hot distilled water until it is free of $AgNO_3$. To the filtrate, add 5 mL of a saturated ferric ammonium sulfate ($FeNH_4(SO_4)_2 \cdot 12H_2O$) solution as an indicator and titrate the solution to a permanent pink tint with standardized 0.1 N ammonium thiocyanate (NH_4SCN) solution. Calculate the weight percent chlorine in the aluminum chloride solution as follows:

Chlorine, wt (weight) percent =

$$3546 \times \frac{(20 \times AgNO_3 \text{ normality}) - (NH_4SCN \text{ normality} \times NH_4SCN \text{ mL added})}{AlCl_3 \text{ sample wt} \times AlCl_3 \text{ wt \%}}$$

Nonconformance to table II shall constitute failure of this test.

4.7.2.3.3 pH of 1 percent aluminum chloride solution. Calculate the weight percent and sample weight of the aluminum chloride solution by performing the test specified in 4.7.2.3.1 and record the values obtained. To a 50-mL volumetric flask pipet X mL of the aluminum chloride solution, where X is equal to the following:

$$X = \frac{2760}{AlCl_3 \text{ sample wt} \times A2cl_3 \text{ wt \%}}$$

Dilute to mark with distilled water. Obtain a standard pH meter, glass electrode, and calomel reference electrode. Calibrate meter with buffer solutions as directed in operator's manual. Measure the pH of the 1 percent $AlCl_3$ solution prepared. Nonconformance to table II shall constitute failure of this test.

4.7.2.4 Trisodium phosphate, anhydrous. Weigh the bag containing the alkaline conditioner to the nearest 0.1 g. Empty the contents to a dry stoppered flask and reweigh the bag. Record the difference in weight as the total weight of conditioner. From the stoppered flask, weigh out 2 g of the conditioner to the nearest 0.1 mg into 500-mL volumetric flask and dilute to mark with distilled water. From this solution pipet 50 mL into a 300-mL Erlenmeyer flask. Dilute to 100 mL with distilled water, then add 10 mL of concentrated nitric acid (HNO_3) and 10 g of ammonium nitrate (NH_4NO_3). On a hot plate heat solution to 30 - 45 °C. Once within the temperature range remove from heat and add 50 mL of ammoniacal molybdate solution (see 4.7.2.4.1). Thoroughly mix solution by shaking or by stirring on a magnetic mixer for 10 minutes. After mixing allow solution to

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sit for 30 minutes to permit the precipitate to settle. After 30 minutes, filter solution through fine textured filter paper, collecting the unwanted filtrate in a beaker. Wash the precipitate and flask with five 20-mL portions of 1 percent potassium nitrate (KNO_3) solution. Similarly wash the filter paper, starting at the rim of the funnel. Return the paper and the precipitate to the 300-mL flask and add standardized 0.5 N sodium hydroxide (NaOH) from a burette until the precipitate decomposes. After decomposition of the precipitate add an excess of 2.0 mL. Record this volume of NaOH added as V_1. Wash the sides of the flask with distilled water. Then dilute to 150 mL with distilled water. Add six drops of phenolphthalein indicator and titrate solution with standardized 0.1 N HNO_3 until pink color disappears. Record the volume of HNO_3 added as V_3. Now titrate the solution back to the previous pink color with standardized 0.1 N NaOH. Record this volume of NaOH added as V_2. Calculate the weight of trisodium phosphate, anhydrous as follows:

$$Na_3PO_4\ wt.\ grams = \frac{0.071291A}{B}\ X\ (V_1N_1 + V_2N_2 - V_3N_3)$$

where:

A = total weight of alkaline conditioner, grams
B = actual weight of 2 g test sample, grams
V_1N_1 = volume added, mL x actual normality of 0.5 N NaOH
V_2N_2 = volume added, mL x actual normality of 0.1 N NaOH
V_3N_3 = volume added, mL x actual normality of 0.1 N HNO_3.

Nonconformance to table III shall constitute failure of this test.

4.7.2.4.1 Preparation of ammoniacal molybdate solution. In a 1-L beaker add 61 g of crystalline ammonium molybdate ($(NH_4)_2MoO_4$), 57 g of ammonium nitrate 47 mL of concentrate ammonium hydroxide (NH_4OH), and 500 mL of distilled water. Warm solution on a hot plate to aid mixing. Allow solution to cool then filter. Transfer solution to a 1-L volumetric flask, dilute to mark with distilled water.

4.7.2.5 Sodium borate, anhydrous. Weigh the bag containing the alkaline conditioner to the nearest 0.1 g. Empty the contents into a dry stoppered flask and reweigh the bag. Record the difference in weight as the total weight of conditioner. From the stoppered flask weigh out 0.9 g of the conditioner to the nearest 0.1 mg into a 300-mL Erlenmeyer flask. Add 75 mL of distilled water. Dissolve conditioner by stirring. Add a five percent solution of barium chloride ($BaCl_2$) in slight excess while stirring. Allow precipitate to settle then filter through a Whatman No. 42 filter paper or equivalent. Wash both the flask

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and the filter paper with two 10-mL portions of distilled water each. Place the filter paper into the flask along with 6 N HCl and dilute mixture to 50 mL. Stir until precipitate has dissolved and the filter paper has fragmented. Add 6 N NaOH until solution is alkaline, indicated by re-precipitation of the phosphate. Add in excess, a few drops each of the $BaCl_2$ solution and the 6 N NaOH to ensure completeness of precipitation. Then filter and wash solution as before. Combine the filtrates and dilute to 200 mL. Make solution slightly acidic by adding 6 N HCl and testing with litmus paper. Add two drops of methyl orange indicator and make solution alkaline again by titrating with standardized 0.5 N NaOH until a clear lemon yellow color is observed. Record this volume added as V_m. After observing yellow color add 1 g of mannitol and 6 drops of phenolphthalein indicator. Titrate solution with the 0.5 N NaOH until a permanent pink end-point is obtained. Add an additional 0.5 g of mannitol and resume the titration until another permanent pink end-point is reached. Record the total volume of NaOH added as V_p. Calculate the weight of sodium borate, anhydrous as follows:

$$Na_2B_4O_7,\ wt\ \% = \frac{(V_p - V_m)\ (5.0307)\ (N)}{weight\ of\ sample}$$

where:

V_p = volume standard NaOH to phenolphthalein end-point
V_m = volume standard NaOH to methyl orange end-point
N = normality of NaOH.

$$Na_2B_4O_7,\ wt,\ grams = \frac{A \times B}{100}$$

where:

A = total weight of alkaline conditioner, grams
B = percent $Na_2B_4O_7$.

Nonconformance to table III shall constitute failure of this test.

4.7.2.6 Sodium carbonate, anhydrous. Weigh the bag containing the alkaline conditioner to the nearest 0.1 g. Empty the contents into a dry stoppered flask and reweigh the bag. Record the difference in weight as the total weight of conditioner. From the stoppered flask weigh out 2.3 g of the conditioner to the nearest 0.1 mg into a 4-mL glass thimble. Place 15 mL of 6 N HCl in a 50-mL flat-bottom flask. With tweezers carefully place thimble filled with conditioner into flask. Stopper flask with a two-holed rubber stopper. Place a drying tube filled with

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calcium chloride ($CaCl_2$) and a glass inlet tube in the holes as shown in figure 1. The drying tube and gas inlet tube should fit snugly. Cover both the drying tube outlet and the gas inlet with rubber policemen. Weigh the entire apparatus to the nearest 0.1 mg, being careful not to overturn the thimble. Remove the rubber policemen from the drying tube outlet and tilt flask to overturn the thimble to initiate the reaction. After the reaction is complete remove the rubber policemen from the gas inlet tube and attach a purification train consisting of a soda lime drying tube and a calcium chloride drying tube. Use vacuum tubing to attach train. Connect a suction line to the calcium chloride drying tube on the flask and apply suction at rate of 3 bubbles per second. Apply suction for 25 minutes, then disconnect suction, replace rubber policemen and reweigh apparatus. Reapply suction for 10 minute time intervals, followed by reweighing, until a constant weight is obtained. Apparatus should experience a net weight loss due to the removal of sodium carbonate (Na_2CO_3). Calculate the weight of sodium carbonate, anhydrous as follows:

$$Na_2CO_3,\ wt\ \% = \frac{(weight\ loss)\ x\ (240.86)}{weight\ of\ sample}$$

$$Na_2CO_3,\ wt = \frac{(Total\ weight\ of\ conditioner)\ x\ (Na_2CO_3,\ wt\ \%)}{100}$$

Nonconformance to table III shall constitute failure of this test.

4.7.2.7 Sodium silicate solution. Weigh the bottle containing the sodium silicate solution to the nearest 0.1 g. Transfer the contents to a dry stoppered flask. Rinse the bottle five times with distilled water and once with 95 percent ethanol. Dry bottle with a stream of air and reweigh bottle. Record the difference in weight as the total weight of the sodium silicate solution. Nonconformance to table III shall constitute failure of this test.

4.7.2.8 Conditioner chemical requirements. Weigh out 10 grams of the alkaline conditioner, to the nearest 0.01 g, into a 125-mL glass stoppered Erlenmeyer flask. Filter 60 mL absolute ethanol through two Whatman No. 41 or equivalent filter papers. With a volumetric pipette, add 50 mL of the filtered ethanol to the flask and shake vigorously for 2 minutes. Pipet 20 mL of this solution into a 120-mL Erlenmeyer flask. Add 20 mL of distilled water and 6 drops of phenolphthalein and titrate with standardized 0.1 N HCl until the pink color disappears. Record the volume obtained. Adjust the volume for HCl concentrations

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other than 0.1 N and report this value as the free alkali in milliliters of 0.1 N HCl. Nonconformance to table IV shall constitute failure of this test.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of packing.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Examination. Each unit of product shall be examined for the following defects.

106. Preservation not as specified for level A (see 5.1).
107. Packing not as specified for level A, B, or C (see 5.2).
108. Marking incorrect, illegible or missing (see 5.3).
109. Instructions for use sheet missing or not as specified (see 5.3.1).
110. Components markings missing or not as specified (see 5.3.2).

5. PACKAGING

5.1 Preservation. Preservation shall level A, B or C as specified (see 6.2).

5.1.1 Level A. The dry components, oxalic acid and alkaline conditioner shall be placed into heat sealable waterproof bags conforming to MIL-B-117, type II, class b. The liquid components, aluminum chloride and sodium silicate, shall be placed in semi-rigid, round polyethylene bottles made of plastic conforming to MIL-B-26701. The cap shall have a self-shrinking cellulose band extending down the neck of the bottle as indicated in figure 3. All components (see figure 3) and instruction sheet (see figure 2) shall be placed into a fiberboard box conforming to PPP-B-636, class 2, W6. Unless otherwise specified, closure shall be in accordance with packaging is method III of MIL-P-116.

5.1.2 Level B. The unit package shall be as specified in 5.1.1.

5.1.3 Level C. The unit packaging shall afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The contractor's standard practice may be used, when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The unit package as described in 5.1.1 shall be packed into a nailed wood box, wood-cleated plywood box, wirebound wood box, conforming to PPP-B-621, class 2; PPP-B-601,

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overseas type; PPP-B-585, class 3. Boxes shall be closed and strapped in accordance with the applicable box specification. Unless otherwise specified, gross weight shall not exceed 200 pounds for wood boxes.

5.2.2 Level B. The unit package as described in 5.1.2 shall be packed into a nailed wood box, wood-cleated plywood box, wirebound wood box, wood-cleated class 1; PPP-B-601, domestic type; PPP-B-585, class 1: PPP-B-591, domestic applicable box specification. Unless otherwise specified, gross weight shall not exceed 200 pounds for wood boxes and 90 pounds for fiberboard boxes.

5.2.3 Level C. The unit package as described in 5.1.3 shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies, and conform to the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

5.3.1 Instructions for use. The instructions for use shall be printed on both sides, if needed, on a 8 by 10-1/2 inch sheet of white bond paper. All printing shall be in capital letters with the underline words in 12 point type and all other in 10 point type. Wording for instructions shall be as shown in figure 2.

5.3.2 Component marking. Each component of the kit shall be marked as indicated in figure 3.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit covered by this specification is intended for use in cooling systems of liquid-cooled internal combustion engines. It has been found that the conditioning treatment of silicate protecting and alkaline conditioning is essential following the cleaning process in order to deactivate the clean metal surface and to rid the system of corrosion products not removed by flushing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required (see 3.1).
d. Level of preservation and packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item(s) should be an initial production sample. The first article should consist of three kits. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirements for samples for the first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Material safety. The contracting officer will ensure that the contractor prepares Material Safety Data Sheets in accordance with FED-STD-313 and that copies are provided to the pertinent Government mailing address that are listed in FED-STD-313, appendix B.

6.5 Disposability. If the components of the product become waste, contact a licensed contractor for detailed recommendations. Follow Federal, state and local regulations.

6.6 Subject term (key word) listing.

Antifreeze
Antifreeze corrosion test kit
Engine coolant system
Hazardous materials
Military coolant
Radiator flush

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6850-1062

Review activities:
Army - AT, SM

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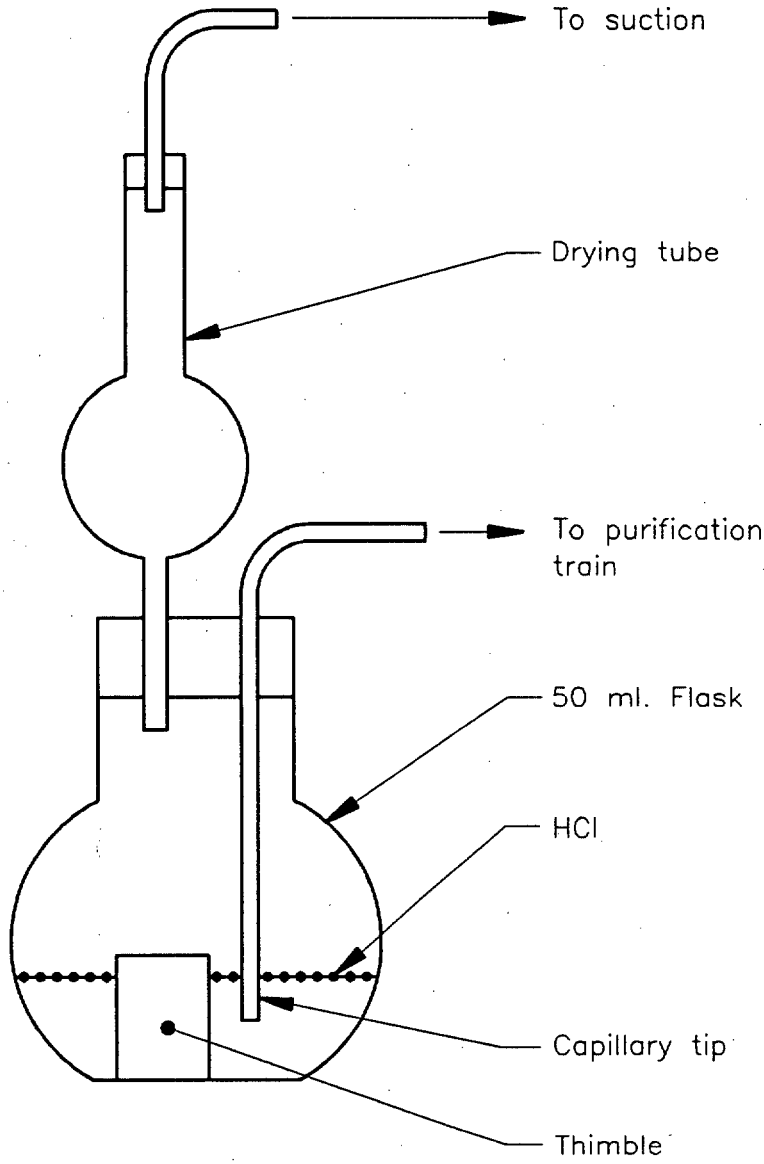


FIGURE 1. Apparatus for testing sodium carbonate, anhydrous.

X-5049

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INSTRUCTIONS FOR USE

COOLING SYSTEM MUST BE CLEANED, CONDITIONED, AND INHIBITED

STEP 1 - CLEANING PROCESS.

DRAIN CONTENTS OF COOLING SYSTEM COMPLETELY. FILL WITH CLEAN WATER TO WITHIN 2 INCHES OF OVERFLOW. IDLE ENGINE UNTIL TEMPERATURE REACHES 160-180 °F. POUR SLOWLY CONTENTS OF BAG MARKED "OXALIC ACID" INTO RADIATOR. POUR CONTENTS OF BOTTLE MARKED "ALUMINUM CHLORIDE" INTO RADIATOR. IDLE ENGINE 20-25 MINUTES. BEST RESULTS ARE OBTAINED IF CLEANING TIME IS NO LONGER THAN 25 MINUTES.

USE: ONE KIT FOR 11 TO 22 QUART COOLING SYSTEM.
TWO KITS FOR 23 TO 44 QUART COOLING SYSTEM.
THREE KITS FOR 45 TO 66 QUART COOLING SYSTEM.

FLUSHING: OPEN ALL DRAIN COCKS AND REMOVE ENGINE BLOCK DRAIN. FLUSH CONTINUALLY, FLOODING COOLING SYSTEM WITH CLEAN WATER. KEEP ENGINE RUNNING FOR 25 MINUTES.

CAUTION: CLEANING PROCESS MUST BE COMPLETED BEFORE CONDITIONING PROCESS IN BEGUN.

STEP 2 - CONDITIONING PROCESS

THE CONDITIONER CONSISTS OF A LIQUID SODIUM SILICATE AND A DRY POWDER ALKALINE CONDITIONER.

SILICATE PROTECTING: FILL COOLING SYSTEM WITH WATER. IDLE ENGINE UNTIL TEMPERATURE REACHES 160-180 °F. SLOWLY ADD CONTENTS OF BOTTLE MARKED "SODIUM SILICATE". IDLE ENGINE FOR 15 MINUTES. DRAIN SYSTEM COMPLETELY.

USE: TWO BOTTLES FOR COOLING SYSTEMS OF 22 TO 44 QUARTS.
THREE BOTTLES FOR COOLING SYSTEMS OVER 44 QUARTS.

ALKALINE CONDITIONING: FILL A 5 GALLON CAN TO WITHIN THREE QUARTERS INCH FROM THE TOP. ADD POWDER FROM BAG MARKED "ALKALINE CONDITIONER". EITHER STIR UNTIL DISSOLVED OR CLOSE CAN TIGHTLY AND IMMEDIATELY ROLL CAN BACK AND FORTH FOR 5 MINUTES. FILL SYSTEM WITH THIS SOLUTION. FOR OTHER SYSTEMS USE IN THIS PROPORTION OF 1 BAG FOR EACH 5 GALLONS OF WATER.

NOTE: SYSTEM OF THE FOLLOWING CAPACITIES MAY BE FILLED WITH WATER, IDLED UNTIL THE TEMPERATURE REACHES 160-180 °F AND THE DRY CONDITIONER ADDED DIRECTLY:

FOR 18-22 QUARTS - ADD DRY CONDITIONER FROM ONE KIT.
36-44 QUARTS - ADD DRY CONDITIONER FROM TWO KITS.
56-66 QUARTS - ADD DRY CONDITIONER FROM THREE KITS.

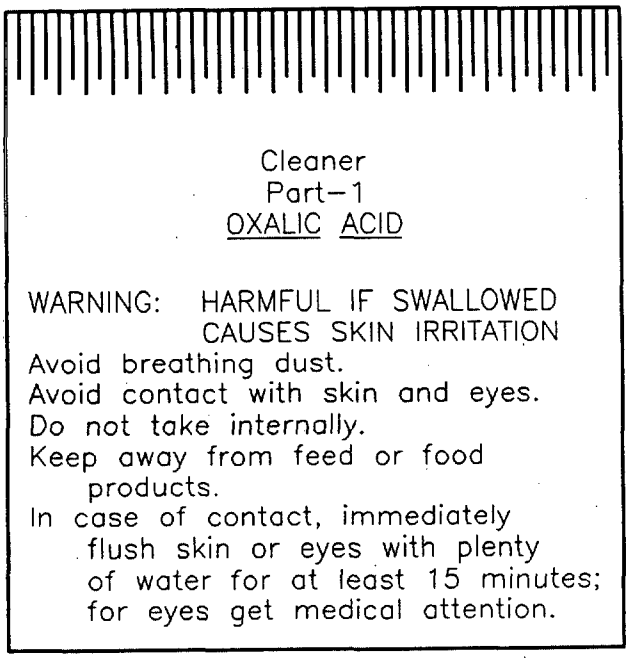
IDLE ENGINE FOR 55-60 MINUTES AFTER REACHING 160-180 °F. DRAIN SYSTEM COMPLETELY. REFILL SYSTEM WITH CLEAN WATER. IDLE ENGINE FOR 15 MINUTES. DRAIN COMPLETELY.

STEP 3 - REFILLING COOLING SYSTEM

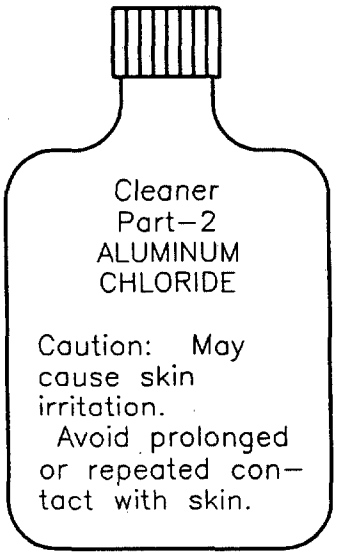
AFTER ENGINE HAS COOLED DRAIN CLEAN WATER. FILL CLEANED AND CONDITIONED SYSTEM WITH MIL-A-46153, MILITARY SPECIFICATION ANTIFREEZE, ETHYLENE GLYCOL, INHIBITED, HEAVY DUTY, SINGLE PACKAGE (NSN 6850-00-181-7929, 6850-00-181-7933, OR 6850-00-181-7940) MIXED 50/50 WITH WATER. IN CLIMATES WHERE TEMPERATURE DOES NOT FALL BELOW +32 °F (0 °C) AND WHERE FREEZE PROTECTION IS NOT REQUIRED, WATER TREATED WITH MIL-A-53009, MILITARY SPECIFICATION ADDITIVE, ANTIFREEZE EXTENDER, LIQUID COOLING SYSTEMS (NSN 6850-01-160-3868) MAY BE USED. THE PROPORTION OF THE MIXTURE SHOULD BE 3% BY VOLUME (E.G., 1 PINT OF EXTENDER TO 17 QUARTS OF WATER).

FIGURE 2. Instruction for use.

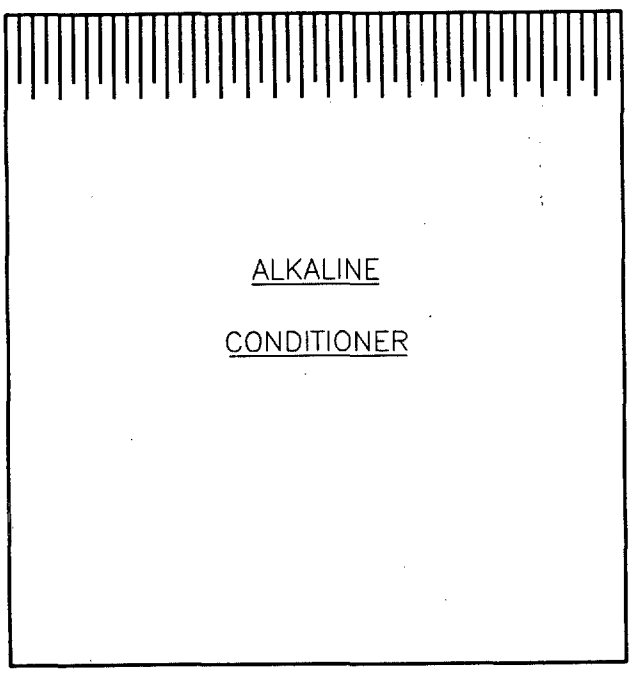
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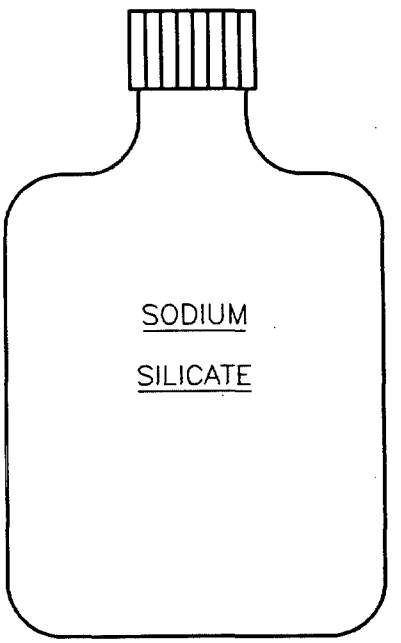
7 x 8



6 oz.



7 x 10



8 oz.

NOTES: BOTTLES SHALL BE PLACED INTO UNIT KIT IN UPRIGHT POSITION INSIDE DIMENSIONS OF BOX, PPP–B–636 FOR UNIT KIT IS 6x6x6

FIGURE 3. Preservation.

X–5050

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-0010597F(ME)	2. DOCUMENT DATE (YYMMDD) 940627

3. DOCUMENT TITLE
CLEANING COMPOUND WITH CONDITIONER FOR ENGINE COOLING SYSTEMS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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